

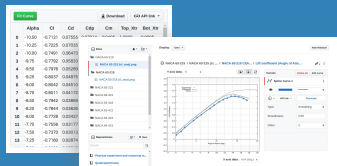


Background:

Spline fitting is widely used to model experimental and simulation data and offers smooth, accurate approximations. However, fitted models are often locked in static plots or scripts, making them difficult to reuse, reproduce, or share across teams and projects.

Our approach:

SplineCloud is an open platform that combines the powers of interactive spline fitting with reusability of models in computer code, offering open API and client libraries.



The Platform:

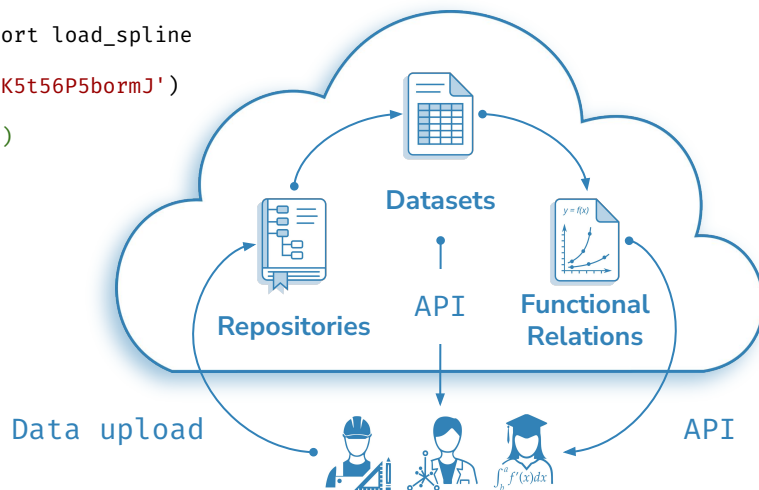
- Has a **repository-based** hierarchical data structure and supports metadata;
- Uses **SciPy spline fitting** methods (UnivariateSpline and LSQUnivariateSpline) on backend and inside client libraries;
- Uses **D3.js** and **verb.js** to support **interactivity** on the frontend;
- Has integrated **plot digitizing tool** to enable fitting curves to plot data;
- Provides **interactive** approach for **editing control points** and **knot vectors** of spline models;
- Makes fitted models **sharable** and **accessible in code** via unique IDs and API links

We built an open platform for fitting and sharing reusable regression models using SciPy



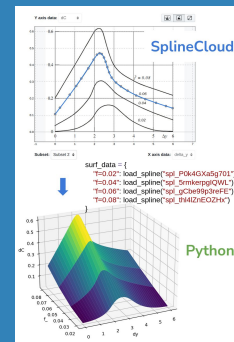
Learn how SplineCloud applies
FAIR principles to curve fitting

```
pip install splinecloud-sciipy  
  
from splinecloud_sciipy import load_spline  
  
spline = load_spline('spl_K5t56P5bormJ')  
  
X = np.linspace(0, 20, 200)  
Y = spline.eval(X)
```



Reusable Curve Fitting Workflow

1. **Create data repository** and make it discoverable: add short description, topics and wiki
2. **Upload data:** spreadsheets, tables, images containing graphs
3. **Define datasets:** create subsets of data or use plot digitizing tool to extract data from graphs
4. **Fit functional relations** using interactive spline fitting tool
5. **Reuse datasets and curves** in code over API and with a help of client libraries



- ✓ Improved reusability of regression models
- ✓ Simplified data processing workflows
- ✓ Improved reproducibility of research

The proposed approach encourages sharing and reuse of regression models across teams and projects. It supports better collaboration, faster modeling workflows, long-term knowledge retention, and more transparent, reproducible research practices.